

"Marlin EV: Front brake pad and rotor service for premature wear from regen

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Area: bulletins

1 Condition

Owners of affected Marlin EV vehicles may report a low-frequency groan or visible scoring of the front rotors well before the expected service interval. Inspection commonly reveals uneven deposit transfer and lipping on the BR-7731 front rotors, accompanied by glazed BP-7720 front brake pads. The condition is most pronounced on vehicles operated primarily in low regenerative-braking modes where the friction brakes carry a disproportionate share of deceleration. See the Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications for the in-service wear limits used to confirm this condition.

2 Cause

The regenerative-to-friction blending calibration on early production units biases too heavily toward friction braking during moderate stops, causing the front pads to run hotter than designed. Sustained elevated temperatures glaze the friction material and accelerate rotor deposit transfer. A revised blending calibration corrects the torque distribution, but vehicles already exhibiting wear require mechanical service. Background on the front brake architecture is documented in the Marlin EV Brakes Service Manual :: Front Brakes section.

3 Repair Procedure

Replace both front BP-7720 pad sets and both BR-7731 rotors as an axle set, following PROC-BRAKE-FR in full. Do not resurface the BR-7731 rotors; the deposit transfer penetrates below the serviceable machining limit and resurfaced rotors will re-glaze. Bed in the new BP-7720 pads per the burnishing schedule and confirm the updated blending calibration is installed before release. Step-by-step torque and bedding guidance is provided in the Marlin EV Brakes Service Manual :: Front Brake Pad and Rotor Service Procedure , and the overarching procedure reference is the Marlin EV Brakes Service Manual .

4 Parts Information

Order the BP-7720 front pad set and the BR-7731 front rotor in pairs to service the front axle as a set. The BP-7720 pad kit includes the wear-sensor clips and anti-rattle shims required for reassembly. Always verify the current part supersession in the catalog before ordering, as the BR-7731 rotor has a revised friction-ring specification on later production. Refer to the Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications for matching dimensions and the full Marlin EV Brakes Service Manual for related fastener part numbers.